

David Nordström

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SUMMARY

- **Research Interests:** 3D Computer Vision (3DV), Self-Supervised Learning (SSL), and Geometric Deep Learning (GDL).
- **Publications:** Main conference papers at ECCV (3), CVPR (1), and ICML (1), whereof 2 orals and 1 spotlight.

EDUCATION

- **Chalmers University of Technology** Gothenburg, Sweden
Ph.D. in Deep Learning Oct. 2024 – Oct. 2028
Description: Supervised by Fredrik Kahl and Georg Bökman. TA in Computer Vision (EEN020) and Deep Learning (SSY340).
- **Seoul National University** Seoul, Korea
Exchange Student Feb. 2023 – Jun. 2023
Description: Visited during the last part of my Master's. Took courses in ML, Math, History, and Korean.
- **University of California, Berkeley** Berkeley, USA
Exchange Student Jun. 2022 – Aug. 2022
Description: Joined as a structured exchange program and took courses in Mathematics (GPA 4.0/4.0).
- **Chalmers University of Technology** Gothenburg, Sweden
M.Sc. in Engineering Mathematics Aug. 2021 – Jun. 2023
Description: Master thesis about $E(3)$ -equivariant networks for molecular dynamics.
- **Chalmers University of Technology** Gothenburg, Sweden
B.Sc. in Industrial Engineering Aug. 2018 – Jun. 2021
Description: GPA 5.0/5.0 and graduated with another B.Sc. in Economics.

EXPERIENCE

- **Ericsson Research** Stockholm, Sweden
Computer Vision Research Intern Mar. 2026 – Jul. 2026
 - **Visual Relocalization with 3DGS:** Worked on developing methods for estimating 6DoF pose for a visual query relative to a map of 3D Gaussian Splats. My work led to a paper and a patent applications.
- **Chalmers University of Technology** Gothenburg, Sweden
Graduate Researcher Oct. 2024 – Present
 - **3D Vision:** Developed state-of-the-art feature extractor outperforming DINOv3 and CroCo v2 on reconstruction, matching, and pose estimation (MuM). Co-created RoMa v2, the current best dense matcher across many tasks and LoMa, the best sparse matcher, outperforming LightGlue by double digits on Visual Localization and Image Matching.
 - **Equivariance & Efficient ViTs:** Demonstrated that horizontal mirroring equivariance reduces FLOPs with no loss in performance. Extended to roto-reflections (5× FLOPs saved), improving DINOv2 and DeiT III. Built custom Triton kernels for high throughput.
- **Kliva** Remote
AI Engineer Aug. 2024 – Present
 - **SWE:** Built cashflow engine in Python, corresponding frontend in React, and Neo4j database connection.
 - **Computer Vision:** Worked on extracting geometry from zoning plans and mapping them to global coordinates.
 - **LLM:** Responsible for building both frontend and backend for a chat interface using an MCP server.
- **Nrep** Stockholm, Sweden
Investment Analyst Aug. 2023 – Aug. 2024
 - **Internship:** Joined as an intern at one of Northern Europe's largest Private Equity firms and received full-time offer.
 - **Full-time:** Summarily, I was part of 12 private deals totaling EUR 200m and was primarily responsible for modeling.

PUBLICATIONS

- **LoMa: Local Feature Matching Revisited:** D. Nordström*, J. Edstedt, G. Bökman, J. Astermark, A. Heyden, V. Larsson, M. Wadenbäck, M. Felsberg, F. Kahl, **Oral** (0.8%) at ECCV 2026.
- **Octic Vision Transformers: Quicker ViTs Through Equivariance:** D. Nordström*, J. Edstedt, F. Kahl, G. Bökman, ECCV 2026.
- **RoMa v2: Harder Better Faster Denser Feature Matching:** J. Edstedt, D. Nordström*, Y. Zhang, G. Bökman, J. Astermark, V. Larsson, A. Heyden, F. Kahl, M. Wadenbäck, M. Felsberg, **Oral** (0.8%) at ECCV 2026.
- **MuM: Multi-View Masked Image Modeling for 3D Vision:** D. Nordström*, J. Edstedt, F. Kahl, G. Bökman, 2026. CVPR 2026.
- **Who Handles Orientation? Investigating Invariance in Feature Matching:** D. Nordström*, J. Edstedt, F. Kahl, G. Bökman. CVPR 2026 Workshop on Image Matching.
- **Flopping for FLOPs: Leveraging equivariance for computational efficiency:** G. Bökman, D. Nordström*, F. Kahl. **Spotlight** (2.6%) at ICML 2025.